

11 (**short**) Figure 10 shows a rotor with moment of inertia J supported by a light elastic shaft of torsional stiffness k . The angle of rotation of the rotor from its equilibrium position is θ .

(a) Determine the natural frequency for torsional oscillations about equilibrium for $J = 1 \text{ kg m}^2$ and $k = 10^4 \text{ N m rad}^{-1}$. [2]

(b) If $\theta = 0.1 \text{ rad}$ and $\dot{\theta} = 10 \text{ rad s}^{-1}$ at $t = 0$, derive an expression for the amplitude of the subsequent motion in the absence of damping. [3]

(c) If the dimensionless damping factor for the system is $\zeta = 0.01$, estimate how long it would take for the amplitude to decrease to 0.1% of its initial value. [5]

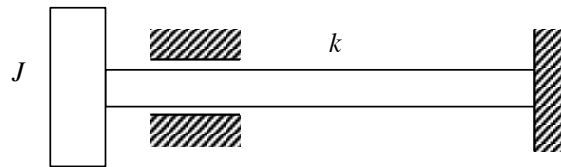


Fig. 10